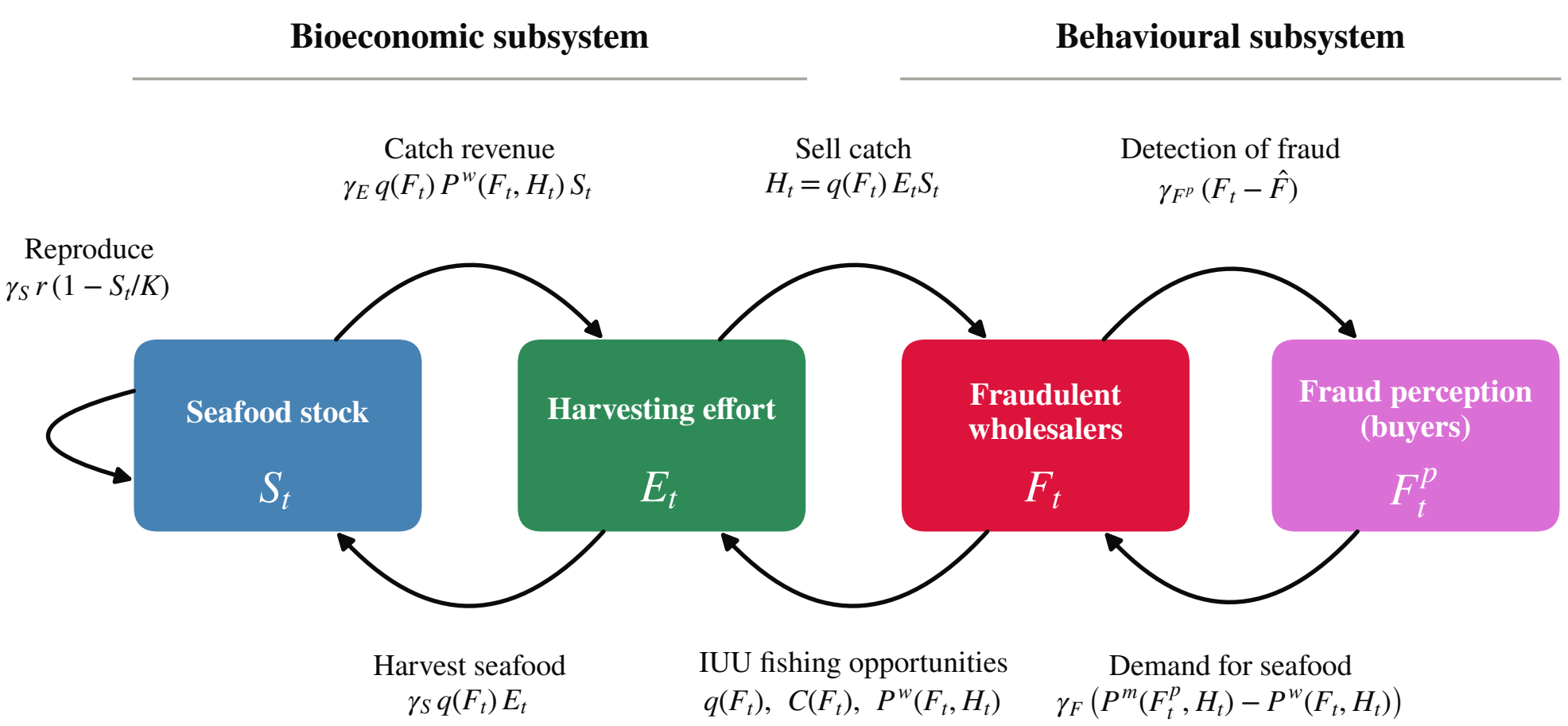
Bioeconomic subsystem
Behavioural subsystem
Catch revenue
$\gamma_E\, q(F_t)\, P^w(F_t, H_t)\, S_t$
Sell catch
$H_t = q(F_t)\, E_t S_t$
Detection of fraud
$\gamma_{F^p}\, (F_t - \hat{F})$
Reproduce
$\gamma_S\, r\, (1 - S_t/K)$
Seafood stock
S_t
Harvesting effort
E_t
Fraudulent wholesalers
F_t
Fraud perception (buyers)
F_t^p
Harvest seafood
$\gamma_S\, q(F_t)\, E_t$
IUU fishing opportunities
$q(F_t),\ C(F_t),\ P^w(F_t, H_t)$
Demand for seafood
$\gamma_F\, \left(P^m(F_t^p, H_t) - P^w(F_t, H_t)\right)$